

Supraorbital versus Infraorbital Transcutaneous Electrical Nerve Stimulation for Migraine Prevention: A Single-Blind, Sham-Controlled Randomized Trial

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Abstract

Background: Migraine is a leading cause of neurological disability. Non-invasive neuromodulation with transcutaneous electrical nerve stimulation (TENS) is a promising preventive option, but randomized head-to-head comparisons of supraorbital versus infraorbital stimulation are scarce.

Objective: To compare the preventive efficacy of bilateral supraorbital versus infraorbital TENS against sham stimulation in adults with migraine.

Methods: In this single-blind, three-arm randomized controlled trial, 83 adults with migraine (ICHD-3) were allocated 1:1:1 to supraorbital TENS (n = 28), infraorbital TENS (n = 27), or sham TENS (n = 28). Active stimulation was delivered bilaterally in clinic (100 Hz; pulse width 200 us; intensity titrated to a strong but tolerable sensory level, target approx. 10 mA; 30 min/session; six sessions/week for 4 weeks). Sham electrodes were placed over the supraorbital nerves without current delivery. The primary outcome was monthly migraine attack frequency at 3 months. Secondary outcomes were pain intensity (VAS), disability (MIDAS), headache impact (HIT-6), oral analgesic-use frequency, and Migraine Prevention Questionnaire (MPQ-5) score, assessed at baseline, 1 month, and 3 months. The trial is reported in accordance with CONSORT 2010 and the CONSORT extension for non-pharmacological treatments.

Results: All 83 randomized participants completed follow-up. Both active interventions produced significant within-group reductions in monthly attack frequency (Friedman $p < 0.001$), whereas sham did not ($p = 0.381$). At 3 months, both active groups were superior to sham on the primary outcome (Kruskal-Wallis $p < 0.001$; supraorbital-sham mean difference -6.35 attacks/month, 95% CI -8.46 to -4.24; infraorbital-sham -6.78, 95% CI -8.86 to -4.70). No statistically significant difference was detected between supraorbital and infraorbital stimulation (mean difference 0.43, 95% CI -0.94 to 1.80). Parallel superiority of active TENS over sham was observed for all secondary outcomes (all Kruskal-Wallis $p < 0.001$; Bonferroni-adjusted $\alpha = 0.01$). Pain intensity decreased by 46.3% and 44.8% with supraorbital and infraorbital TENS, respectively, versus 8.0% with sham. No serious adverse events occurred.

Conclusions: In this selected, highly adherent outpatient cohort, supraorbital and infraorbital TENS each improved migraine-related outcomes relative to sham. Between-site differences were not demonstrated. Given underpowering for the site comparison, sham-site asymmetry, incomplete participant blinding, and pre-randomization enrichment for schedule adherence, these findings should be regarded as preliminary and require confirmation in larger, site-matched, more rigorously blinded trials.

Trial registration: Ethics approval code: [TO BE COMPLETED]; IRCT registration number: [TO BE COMPLETED].

Keywords: Migraine; Transcutaneous electrical nerve stimulation; Supraorbital nerve; Infraorbital nerve; Trigeminal neuromodulation; Randomized controlled trial

1. Introduction

Migraine is a prevalent and disabling neurological disorder and remains among the leading causes of years lived with disability worldwide [1,34]. It is characterized by recurrent attacks of moderate-to-severe headache, typically lasting 4-72 hours and often accompanied by photophobia, phonophobia, nausea, or vomiting [2]. Contemporary models conceptualize migraine as a primary brain disorder involving activation and sensitization of the trigeminovascular pathway, with altered connectivity among cranio-cervical afferents, brainstem nuclei, and higher-order networks [3,28].

Migraine imposes a substantial clinical and socioeconomic burden. Headache accounts for a sizeable fraction of primary-care and neurology outpatient visits [4-6], and a minority of patients develop high-frequency or chronic migraine [7,8], amplifying disability and healthcare utilization [9]. Although

pharmacological prophylaxis and acute therapies are central to management, adverse effects, incomplete response, and medication-overuse headache frequently limit long-term benefit [10,35]. These constraints have intensified interest in non-pharmacological strategies with favorable tolerability.

Peripheral neuromodulation is one such approach. Transcutaneous electrical nerve stimulation (TENS) delivers pulsed currents across intact skin to modulate nociceptive signaling [11-13]. Gate-control and descending-inhibition frameworks provide a mechanistic rationale for analgesic effects [13,14]. The Cefaly device, which stimulates the supraorbital and supratrochlear nerves, was the first FDA-cleared non-invasive device for migraine prophylaxis [14,15]. Subsequent studies of supraorbital TENS (external trigeminal nerve stimulation; eTNS/tSNS) have reported reductions in attack frequency, headache days, and analgesic use, with generally mild adverse effects [15-21]. Stimulation parameters themselves may influence clinical response [22].

Evidence for infraorbital nerve stimulation is far more limited and largely derived from nerve-block or implanted-stimulator series rather than randomized TENS trials [23,24]. Because both the ophthalmic and maxillary divisions converge on shared trigeminocervical circuits, a direct comparison of supraorbital and infraorbital surface stimulation is clinically and scientifically relevant. We therefore conducted a randomized, sham-controlled trial to compare bilateral supraorbital versus infraorbital TENS on monthly migraine attack frequency (primary outcome) and on pain intensity, disability, headache impact, and medication-related secondary outcomes.

2. Methods

2.1. Study design and setting

We conducted a single-blind, three-arm randomized controlled trial (2020-2024) in the outpatient Physical Medicine and Rehabilitation clinics of Isfahan University of Medical Sciences, Iran. Reporting follows the CONSORT 2010 statement and the CONSORT extension for non-pharmacological treatments. The protocol was approved by the institutional ethics committee (approval code: [TO BE COMPLETED]) and prospectively registered in the Iranian Registry of Clinical Trials (IRCT: [TO BE COMPLETED]). All participants provided written informed consent, and the study adhered to the Declaration of Helsinki.

2.2. Participants

Eligible participants were adults aged 18-65 years with migraine with or without aura according to the International Classification of Headache Disorders, 3rd edition (ICHD-3; codes 1.1 and 1.2.1) [2], and ≥ 2 migraine attacks per month during the preceding three months. Monthly migraine attack frequency was defined as the number of discrete migraine attacks (not headache days), ascertained from a prospective diary; a new attack was counted after a pain-free interval of ≥ 24 hours.

Of 137 patients assessed for eligibility, 54 were excluded before randomization because they were unable or unwilling to commit to the intensive schedule (six clinic sessions weekly for four weeks; 24 sessions). The remaining 83 patients were randomized (Figure 1). Consequently, the randomized sample was enriched for individuals with high schedule flexibility and treatment commitment. This selection bias is acknowledged as a limitation affecting external validity (Section 4.7) and most plausibly explains the absence of post-randomization attrition.

2.3. Exclusion criteria

Exclusion criteria were: prophylactic migraine medication within two weeks before enrollment; preventive interventions (e.g., nerve block, botulinum toxin) within the past year; medication-overuse headache (ICHD-3 8.2); other concurrent primary headache disorders requiring active management; severe psychiatric or systemic illness; pregnancy or lactation; substance abuse; and anticipated inability to comply with the protocol.

2.4. Sample size

The original dichotomous responder-rate calculation was incongruent with the continuous primary endpoint; the sample size was therefore recalculated for monthly attack frequency. Using PREMICE trial data [15], an assumed SD of approx. 4.5 attacks, 80% power, two-sided $\alpha = 0.05$, and a minimum clinically important between-group difference of 3 attacks/month yielded approx. 35 participants per group.

Allowing 20% attrition, the ideal target was 42 per group (approx. 126 total). Because of feasibility constraints, a practical target of ≥ 27 per group was adopted (planned n approx. 81); 83 participants were ultimately randomized (28, 27, and 28). Achieved power for the planned 3-attack contrast was approx. 64%. Post-hoc power for the observed active-versus-sham contrasts exceeded 99% because of large observed effect sizes; this estimate is descriptive only and does not validate the a priori design.

2.5. Randomization and blinding

Participants were allocated 1:1:1 using computer-generated sequences in SPSS version 22 (IBM Corp., Armonk, NY, USA) with a block size of 4. Allocation was concealed with sequentially numbered, sealed, opaque envelopes. Session schedules and electrode procedures were matched across arms insofar as the design allowed.

Because the sham arm delivered no perceptible current, complete sensory blinding cannot be assumed. A post-intervention blinding assessment asked participants to guess allocation: 62% in the active arms and 54% in the sham arm guessed correctly ($p = 0.34$). Outcome assessment and statistical analysis were performed by an investigator blinded to group assignment.

2.6. Interventions

Supraorbital TENS: Bilateral surface stimulation over the supraorbital foramina/notches under therapist supervision, using a Novin Co. (Iran) TENS device. Parameters: 100 Hz; pulse width 200 μ s; 30 min/session; six sessions/week for 4 weeks (24 planned sessions). Intensity was titrated to a strong but tolerable paresthesia (target approx. 10 mA).

Infraorbital TENS: Identical parameters, schedule, and device applied bilaterally over the infraorbital foramina.

Sham TENS: Electrodes were placed over the supraorbital nerves with identical appearance, schedule, and session duration; the stimulator was switched on visually but delivered no current (inert sham). The sham condition therefore replicated only the supraorbital montage; the infraorbital-versus-sham contrast is not site-matched (Section 4.7).

All participants continued usual acute migraine therapy (acetaminophen, NSAIDs, or ergot derivatives) as needed. Adherence was defined as attendance at scheduled clinic sessions; all randomized participants completed the planned schedule.

2.7. Outcomes

Outcomes were assessed at baseline, 1 month, and 3 months after treatment initiation.

Primary outcome: Monthly migraine attack frequency at 3 months.

Secondary outcomes: (i) pain intensity on a 0-10 visual analog scale (VAS); (ii) Migraine Disability Assessment (MIDAS) [33]; (iii) Headache Impact Test (HIT-6) [32,38]; (iv) frequency of oral analgesic use (occasions/month); and (v) Migraine Prevention Questionnaire (MPQ-5) score [31]. Because five secondary endpoints were examined without a pre-specified hierarchy, a Bonferroni-adjusted threshold of $\alpha = 0.01$ ($0.05/5$) was applied for secondary inferences.

Adverse events were captured by open-ended clinician inquiry at each visit rather than a standardized checklist.

2.8. Statistical analysis

Analyses used SPSS version 22. Continuous variables are summarized as mean $\pm$ SD. Baseline between-group comparisons used one-way ANOVA for continuous variables and Pearson's chi-square test for categorical variables. Within-group changes across the three time points were evaluated with the Friedman test. Between-group differences at 3 months were assessed with the Kruskal-Wallis test, followed by pairwise Mann-Whitney U tests.

Between-group mean differences at 3 months are accompanied by approximate 95% confidence intervals (CIs) calculated from group means and SDs (Welch formula). These CIs are descriptive effect-size supplements to the non-parametric tests and are not formally aligned with the rank-based inference.

All 83 randomized participants completed assessments and were analyzed as randomized (complete-case; no imputation required). Two-sided $p < 0.05$ defined statistical significance for the primary between-group contrast. We did not perform imputation-based sensitivity analyses that reassign pre-randomization exclusions into a single study arm, because such patients were never randomized and cannot validly be attributed to one post-allocation group; instead, selection bias is addressed qualitatively in the Discussion.

3. Results

3.1. Participants

Of 137 screened patients, 54 were excluded before randomization for inability or unwillingness to complete the 24-session schedule, and 83 were randomized to supraorbital TENS ($n = 28$), infraorbital TENS ($n = 27$), or sham ($n = 28$) (Figure 1). All randomized participants received the allocated intervention, completed the 4-week schedule, and provided 1- and 3-month outcomes; there were no post-randomization dropouts.

Baseline age and sex were balanced across groups (Table 1). Overall, 63/83 participants (75.9%) were female. Mean baseline attack frequency was high (approximately 12-15 attacks/month), indicating a high-frequency migraine cohort, although formal episodic-versus-chronic subgrouping was not pre-specified.

Table 1. Baseline demographic characteristics. Values are mean $\pm$ SD or n. No significant between-group differences were observed.

Variable	Supraorbital ($n=28$)	Infraorbital ($n=27$)	Sham ($n=28$)	p-value
Mean age, years	35.8 $\pm$ 7.6	36.4 $\pm$ 6.9	35.1 $\pm$ 7.2	0.72
Sex (F/M)	21/7	20/7	22/6	0.94

No serious adverse events were recorded. Mild stimulation-related sensations were expected with active TENS and did not lead to discontinuation (Section 3.5).

3.2. Primary outcome: migraine attack frequency

Both active groups showed significant reductions in monthly attack frequency from baseline to 3 months (supraorbital: 14.18 $\pm$ 3.95 to 6.59 $\pm$ 2.70; infraorbital: 15.13 $\pm$ 4.51 to 6.16 $\pm$ 2.48; Friedman $p < 0.001$). The sham group showed no significant change (12.32 $\pm$ 4.99 to 12.94 $\pm$ 5.01; $p = 0.381$) (Table 2; Figure 2).

Table 2. Monthly migraine attack frequency (primary outcome). Data are mean $\pm$ SD; within-group p-values are from Friedman tests.

Group	Baseline	1 month	3 months	p-value
Supraorbital	14.18 $\pm$ 3.95	7.68 $\pm$ 2.84	6.59 $\pm$ 2.70	<0.001
Infraorbital	15.13 $\pm$ 4.51	9.19 $\pm$ 4.03	6.16 $\pm$ 2.48	<0.001
Sham	12.32 $\pm$ 4.99	14.27 $\pm$ 4.15	12.94 $\pm$ 5.01	0.381

3.3. Secondary outcomes

Pain intensity (VAS) decreased significantly in both active groups (supraorbital: 7.30 $\pm$ 1.29 to 3.92 $\pm$ 0.97; infraorbital: 7.07 $\pm$ 1.47 to 3.90 $\pm$ 1.33; Friedman $p < 0.001$) but not in sham (7.10 $\pm$ 1.11 to 6.53 $\pm$ 1.02; $p = 0.125$) (Table 3).

Table 3. Pain intensity (VAS). Data are mean $\pm$ SD.

Group	Baseline	1 month	3 months	p-value
Supraorbital	7.30 $\pm$ 1.29	4.07 $\pm$ 0.76	3.92 $\pm$ 0.97	<0.001
Infraorbital	7.07 $\pm$ 1.47	5.05 $\pm$ 1.24	3.90 $\pm$ 1.33	<0.001
Sham	7.10 $\pm$ 1.11	6.16 $\pm$ 1.44	6.53 $\pm$ 1.02	0.125

MIDAS scores improved in both active groups (supraorbital: 20.02 $\pm$ 5.22 to 9.76 $\pm$ 3.12; infraorbital: 17.94 $\pm$ 6.30 to 10.30 $\pm$ 2.07; $p < 0.001$) but not in sham (19.36 $\pm$ 4.05 to 15.96 $\pm$ 5.39; $p = 0.174$) (Table 4). HIT-6 decreased in both active groups (supraorbital: 62.58 $\pm$ 4.78 to 49.48 $\pm$ 4.84; infraorbital: 64.22 $\pm$ 5.93 to 51.40 $\pm$ 3.01; $p < 0.001$) without significant change in sham (61.11 $\pm$ 3.74

to 62.18 +/- 4.13; p = 0.364) (Table 5).

Table 4. MIDAS disability scores. Data are mean +/- SD.

Group	Baseline	1 month	3 months	p-value
Supraorbital	20.02 +/- 5.22	12.33 +/- 3.94	9.76 +/- 3.12	<0.001
Infraorbital	17.94 +/- 6.30	13.19 +/- 4.62	10.30 +/- 2.07	<0.001
Sham	19.36 +/- 4.05	17.87 +/- 5.20	15.96 +/- 5.39	0.174

Table 5. HIT-6 scores. Data are mean +/- SD.

Group	Baseline	1 month	3 months	p-value
Supraorbital	62.58 +/- 4.78	51.50 +/- 3.61	49.48 +/- 4.84	<0.001
Infraorbital	64.22 +/- 5.93	51.50 +/- 5.39	51.40 +/- 3.01	<0.001
Sham	61.11 +/- 3.74	62.39 +/- 4.20	62.18 +/- 4.13	0.364

Oral analgesic use decreased significantly with active TENS (supraorbital: 14.62 +/- 4.35 to 5.02 +/- 2.64; infraorbital: 14.72 +/- 3.50 to 5.65 +/- 2.55; p < 0.001) but not with sham (14.36 +/- 4.27 to 12.06 +/- 4.25; p = 0.066). MPQ-5 scores likewise decreased in both active groups (p < 0.001) without significant sham change (p = 0.507) (Table 6).

Table 6. Medication-related outcomes. Data are mean +/- SD.

Variable / Group	Baseline	1 month	3 months	p-value
Analgesic use - Supraorbital	14.62 +/- 4.35	6.21 +/- 2.59	5.02 +/- 2.64	<0.001
Analgesic use - Infraorbital	14.72 +/- 3.50	8.57 +/- 3.20	5.65 +/- 2.55	<0.001
Analgesic use - Sham	14.36 +/- 4.27	12.72 +/- 2.88	12.06 +/- 4.25	0.066
MPQ-5 - Supraorbital	26.43 +/- 6.49	16.73 +/- 5.74	16.77 +/- 5.08	<0.001
MPQ-5 - Infraorbital	26.52 +/- 6.23	15.98 +/- 4.63	14.87 +/- 5.34	<0.001
MPQ-5 - Sham	25.42 +/- 6.40	24.17 +/- 6.49	24.16 +/- 4.23	0.507

3.4. Between-group comparisons at 3 months

Kruskal-Wallis tests showed significant between-group differences at 3 months for the primary outcome and all secondary outcomes (all p < 0.001). Pairwise contrasts indicated superiority of both active arms over sham, without statistically significant differences between supraorbital and infraorbital stimulation (Table 7). All secondary between-group contrasts remained significant under the Bonferroni-adjusted alpha = 0.01.

Table 7. Between-group comparisons at 3 months. Overall p-values are from Kruskal-Wallis tests. Mean differences (MD) with approximate 95% CIs use the Welch formula (descriptive). Negative MD favors the first-named group when lower values indicate improvement.

Outcome	KW p	Post-hoc summary (MD [95% CI])
Attack frequency	<0.001	Supra-sham -6.35 (-8.46, -4.24); Infra-sham -6.78 (-8.86, -4.70); Supra-infra 0.43 (-0.94, 1.80)
VAS	<0.001	Supra-sham -2.61 (-3.13, -2.09); Infra-sham -2.63 (-3.26, -2.00); Supra-infra 0.02 (-0.60, 0.64)
MIDAS	<0.001	Supra-sham -6.20 (-8.51, -3.89); Infra-sham -5.66 (-7.80, -3.52); Supra-infra -0.54 (-1.93, 0.85)
HIT-6	<0.001	Supra-sham -12.70 (-15.06, -10.34); Infra-sham -10.78 (-12.69, -8.87); Supra-infra -1.92 (-4.04, 0.20)
Analgesic use	<0.001	Supra-sham -7.04 (-8.89, -5.19); Infra-sham -6.41 (-8.25, -4.57); Supra-infra -0.63 (-2.00, 0.74)
MPQ-5	<0.001	Supra-sham -7.39 (-9.84, -4.94); Infra-sham -9.29 (-11.84, -6.74); Supra-infra 1.90 (-0.86, 4.66)

3.5. Adverse events and tolerability

In the active arms, the most common adverse effects were mild, expected electrode-site sensations recorded by open-ended inquiry: itching in 9/28 (32.1%) supraorbital and 8/27 (29.6%) infraorbital participants; paresthesia/tingling in 12/28 (42.9%) and 11/27 (40.7%), respectively. All events were transient and did not require discontinuation. No serious adverse events occurred in any arm. Open-ended capture may have under-ascertained milder effects.

4. Discussion

4.1. Principal findings

In this single-blind, sham-controlled randomized trial, preventive TENS applied over either the supraorbital or the infraorbital nerve reduced monthly migraine attack frequency and improved secondary patient-reported outcomes relative to sham at 3 months. Between-site differences were not statistically significant. Because the trial was neither powered nor pre-specified for equivalence or non-inferiority, the absence of a site difference must not be interpreted as proven equivalence, particularly given sham-site asymmetry.

The magnitude of change was clinically notable: VAS reductions of approx. 45-46% with active stimulation versus 8% with sham, and HIT-6 reductions exceeding the commonly cited approx. 5-point within-person minimal clinically important difference [39]. Nevertheless, all outcomes were patient-reported, participant blinding was incomplete, and the cohort was enriched for high adherence; expectation effects and selection may therefore have contributed to part of the observed effect sizes alongside any physiological action of TENS.

4.2. Relation to prior work

Our findings are directionally consistent with evidence supporting trigeminal neuromodulation in migraine. The TEAM trial demonstrated benefit of eTNS for acute migraine outcomes [25], and a systematic review of neuromodulation RCTs concluded that selected techniques, including supraorbital TENS for prevention, can improve migraine-related endpoints with a generally favorable safety profile, albeit with limited overall evidence quality and typically small-to-moderate effects [26]. The relative VAS reductions observed here are broadly comparable to responder patterns reported for supraorbital neurostimulation [27], but cross-trial numerical comparisons should be cautious given differences in design, population, and outcome definitions.

4.3. Site comparison: descriptive patterns only

Numerically, supraorbital stimulation showed a faster early VAS trajectory (44.2% reduction at 1 month; 46.3% at 3 months), whereas infraorbital stimulation improved more gradually (28.6% then 44.8%), converging by 3 months (supra-infra MD 0.02, 95% CI -0.60 to 0.64). These patterns were not statistically tested as a pre-specified time-by-site interaction and are hypothesis-generating only. A speculative anatomical account—that the more superficial supraorbital terminal branch may permit earlier afferent recruitment, whereas infraorbital (maxillary) afferents engage partially distinct pathways before converging on shared trigeminocervical circuits [28,29]—remains untested. Broader proposed mechanisms of TENS analgesia include A-beta-mediated gate control, descending inhibition, and monoaminergic/CGRP-related modulation [3,13,28,29]; this trial included no imaging or biomarker measures capable of adjudicating among them.

4.4. Disability, impact, and medication use

MIDAS decreased by 51.2% (supraorbital) and 42.6% (infraorbital). HIT-6 fell from the severe-impact range (≥ 60) to approximately the little-to-no-impact border in the supraorbital arm (49.48) and to "some impact" in the infraorbital arm (51.40) [32]. Oral analgesic frequency decreased by 65.7% and 61.6%, respectively, and MPQ-5 scores also declined with active treatment. Medication-related benefits are clinically relevant given the risks of adverse effects and medication-overuse headache with pharmacological management [10,34,35], although pooled evidence for related non-invasive modalities remains mixed for some medication-day endpoints [40].

4.5. Clinical implications

Pending confirmatory research, either stimulation site may be considered when TENS is used preventively in patients able to attend an intensive in-clinic schedule; practical factors (montage feasibility, tolerability, preference) may guide site choice more than a claimed efficacy advantage. Expert consensus supports neuromodulation for selected patients, particularly those refractory to or intolerant of drugs [36,37]. The present protocol does not establish acute abortive efficacy. Home-based, fewer-session, or remotely supervised regimens will be needed before broader implementation can be recommended.

4.6. Strengths

Strengths include randomized sham control, investigator-blinded outcome assessment and analysis, multidimensional migraine-specific outcomes, balanced baseline demographics for age and sex, and

3-month follow-up. To our knowledge, this is among the first RCTs to compare transcutaneous supraorbital and infraorbital stimulation directly in migraine.

4.7. Limitations and future directions

Several limitations constrain interpretation and generalizability:

1. Pre-randomization selection bias: 54/137 screened patients were excluded solely for inability to commit to 24 sessions. The analyzed sample is enriched for high adherence and does not represent the broader migraine population. We deliberately avoided imputing never-randomized individuals into a single study arm, as that approach is statistically invalid; selection bias is therefore acknowledged rather than "adjusted away."
2. Underpowered site comparison: Achieved power for the planned 3-attack contrast was approx. 64%; the non-significant supraorbital-versus-infraorbital result is inconclusive, not evidence of equivalence.
3. Sham-site asymmetry: Sham replicated only the supraorbital montage; the infraorbital-versus-sham contrast is not site-matched and may partly reflect montage-specific expectancy or sensory cues.
4. Incomplete participant blinding with exclusively patient-reported outcomes raises the possibility that expectation inflated effect estimates.
5. Approximate Welch CIs accompany non-parametric tests and are descriptive rather than model-consistent interval estimates.
6. Follow-up of 3 months does not establish durability.
7. Adverse events were captured by open-ended inquiry rather than a standardized checklist.
8. Single-center design and a predominantly female sample (75.9%) limit external validity.
9. No pre-specified subgroup analyses by migraine subtype (episodic vs chronic; with vs without aura), despite high baseline attack frequency suggestive of a high-frequency cohort.
10. Ethics approval and IRCT numbers remain placeholders and must be inserted before journal submission.

Future trials should use larger samples; site-matched, more credible sham conditions; rigorous blinding with pre-specified blinding indices; longer follow-up; a locked statistical analysis plan; mechanistic imaging or biomarkers; dose-finding designs; and predictors of response. Pragmatic evaluations of lower-burden delivery models are also needed.

5. Conclusions

In a selected, highly adherent outpatient population, preventive TENS over either the supraorbital or infraorbital trigeminal branch improved monthly migraine attack frequency and secondary patient-reported outcomes relative to sham at 3 months. No statistically significant between-site difference was demonstrated; given limited power and sham-site asymmetry, this finding is inconclusive rather than evidence of equivalence. Descriptive early differences in VAS trajectory are hypothesis-generating only. Confirmation in larger, site-matched, more rigorously blinded trials with longer follow-up and verifiable prospective registration is required before strong practice recommendations can be made.

Declarations

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Authors' contributions: All authors contributed to conception and design, data acquisition or interpretation, and critical revision of the manuscript, and approved the final version. Detailed CRediT roles should be finalized before submission.

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Figure legends

Figure 1. CONSORT flow diagram of participant screening, exclusion, randomization, intervention, and follow-up.

Figure 2. Mean monthly migraine attack frequency at baseline, 1 month, and 3 months by study group. Error bars represent SD.

Note to authors: Insert completed CONSORT diagram (Figure 1) and outcome figure (Figure 2). Replace all [TO BE COMPLETED] ethics/IRCT placeholders before submission. Cite reference [30] in the Discussion if retained, or remove it from the reference list.